

PAPER_1308_LEPTON_CP_DELTA

Daniel T. Murphy

PAPER_1308 — Lepton CP Violation $\delta_{CP} = -\pi/2$

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: C — Particle Physics Open (CLOSED)

Date: June 11, 2026

Calculator surface: `calculate_paradox({"paradox": "lepton_cp_violation"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1308})`

Master Expression

``

$\delta_{CP} = -\pi/2 = -90^\circ$ via maximal F_{TRZ} phase lock

``

Match: STRUCTURAL INTEGER

Interpretation

Maximal CP violation in lepton sector via $F_{TRZ} \times \pi/2$ phase. Consistent with T2K/NOvA.

UQFF Locked Primitives

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$

- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$

- $\Lambda_{QCD} = 0.217$ GeV, $v_{Higgs} = 246$ GeV (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 11, 2026, Youngstown, OH.

Subject matter: UQFF derivation of Lepton CP Violation $\delta_{CP} = -\pi/2$.